A preliminary study of notions of independence for choice functions

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Abstract. We study independence concepts for imprecise-probabilistic choice functions. We introduce representation independence and a stronger variant, and we derive their corresponding independent product, which is the smallest coherent choice function that marginalises to given choice functions and satisfies the respective notion of independence. By applying similar proof techniques, we also obtain the independent product for the existing S-independence. We make comparisons between these products, and derive some of their properties. Finally, we discuss open questions concerning the independence notions.

Keywords: Independence · Sets of desirable gamble sets · Choice functions · Natural extension.

1 Introduction

If an agent's joint beliefs about two uncertain variables (X, Y) are described by a probability mass function p on their joint possibility space,[1] then X and Y are independent if conditioning p on any value of one variable does not change the beliefs about the other. This implies that, given marginal mass functions p_X and p_Y, the (unique) independent joint distribution with these marginals is their product $p_X p_Y$. This construction is a form of *independent natural extension*, which we study here in an imprecise-probabilistic setting.

In this paper we investigate independence for choice functions, one of the most expressive belief models in the imprecise-probabilistic framework. Their added flexibility allows expressing alternative notions of independence, such as requiring independence in every model of a representation, or judging it unreasonable to use observations of X to make decisions about Y. We introduce two new notions of independence and relate them to two existing ones.

The paper is organised as follows. In Section 2, we introduce the relevant results about choice functions, or their equivalent model of sets of desirable gamble sets. Section 3 contains the main results: after reviewing marginalisation and conditioning, we introduce the independence notions. Section 4 concludes with a brief discussion and directions for future research. This version is identical to our first submission to IPMU 2026, but with the proofs of all results included in the appendix, to allow the reviewers to verify the claimed results.

[1] For convenience, we assume all probabilities are positive.

2 Choice functions

Consider an uncertain variable X taking values in a possibility space $\mathcal{X}$, which we will assume to be *finite* throughout this paper. A gamble f is a real-valued function on $\mathcal{X}$ that maps each possibility x in $\mathcal{X}$ to a reward $f(x)$, expressed in unites of a predetermined utility scale. If the agent accepts a gamble f, she accepts to receive the – possibly negative – reward $f(x)$ if the possibility x is realised.

Because X is an *uncertain* variable, a gamble f on $\mathcal{X}$ involves risk so it is a risky transaction. An agent's preferences between gambles therefore reveal her beliefs about X. A *choice function* C is a function that describes the agent's preference between gambles in a very general way, without assuming that they are binary. In this paper, we study choice functions as a belief model, but for mathematical convenience, we shall focus on the equivalent model of *sets of desirable gamble sets*.

To introduce choice functions, we first need to introduce some relevant notation. We denote all gambles on $\mathcal{X}$ as $\mathcal{L}(\mathcal{X})$ – or simply $\mathcal{L}$ when it is clear from the context what the domain is. The set of all gambles $\mathcal{L}(\mathcal{X})$ is an $|\mathcal{X}|$-dimensional linear space under the pointwise addition and scalar multiplication of gambles. For two gambles f and g, we use $f \leq g$ to denote $f(x) \leq g(x)$ for all x in $\mathcal{X}$, and we use $f < g$ for '$f \leq g$ and $f(x) < g(x)$ for some x in $\mathcal{X}$'. A constant gamble is a gamble that returns the same value α everywhere, and we identify such gamble with a real number α. We collect the gambles that result in a partial gain in the *set of positive gambles* $\mathcal{L}_{>0}(\mathcal{X}) := \{f \in \mathcal{L}(\mathcal{X}) : f > 0\}$, and the gambles that cannot make you win utility in the *set of non-positive gambles* $\mathcal{L}_{\leq 0}(\mathcal{X}) := \{f \in \mathcal{L}(\mathcal{X}) : f \leq 0\}$ – or simply $\mathcal{L}_{>0}$ and $\mathcal{L}_{\leq 0}$ respectively, when it is clear from the context what the domain is.

Choice functions take a *finite* set of gambles, called *gamble set*, as an argument. We denote the set of all gamble sets on $\mathcal{X}$ as $\mathcal{Q}(\mathcal{X}) := \{F \subseteq \mathcal{L}(\mathcal{X}) : |F| \in \mathbb{N}\}$, or simply $\mathcal{Q}$. A choice function C on $\mathcal{X}$ is a function $\mathcal{Q}(\mathcal{X}) \to \mathcal{Q}(\mathcal{X})$ such that $C(F) \subseteq F$ for every F in $\mathcal{Q}(\mathcal{X})$. The set $C(F)$, called the *choice set*, is the set of non-rejected, or admissible, gambles in F. A choice function C can also be characterised via its associated *rejection function* R, defined by $R(F) := F \setminus C(F)$ for each F in $\mathcal{Q}(\mathcal{X})$. The function R identifies the set $R(F)$ of rejected gambles from F. A choice function C and a rejection function R are equivalent representations of the same information, but it is often more convenient to work with the latter.

Imprecise-probabilistic choice functions have first been introduced by Seidenfeld et al [8,13], where they have provided an axiomatisation for coherent choice between *horse lotteries* rather than gambles. Since then, Refs. [14,3,4,6,15,21], among others, have proposed an axiomatisation based on it, but specified to gambles. Due to page limitations, we will not give an overview of the rationality axioms for choice functions here, but rather show them directly in Section 2.1 in their equivalent and more convenient form – sets of desirable gamble sets – established in Refs. [3,4]. We stress that the rationality axioms allow for a gamble f to be admissible from both $\{f, g\}$ and $\{f, h\}$, while being inadmissible

from $\{f, g, h\}$, which proves that choice functions can express more than pairwise preferences.

2.1 Sets of desirable gamble sets

In this section, we will discuss relevant results about coherent sets of desirable gamble sets, which were introduced in Refs. [3,4] as an equivalent representation of coherent choice functions. Given the equivalence, everything we establish for sets of desirable gamble sets, will immediately hold for choice functions too, *mutatis mutandis*.

Definition 1 (Sets of desirable gamble sets). *A set of desirable gamble sets K on $\mathcal{X}$ is a subset of $\mathcal{Q}(\mathcal{X})$.*

The idea is that a set of desirable gamble sets K contains all the gamble sets F that have at least one gamble that the agent prefers to the constant gamble 0, called the *status quo*. Such a gamble set F is then called a *desirable gamble set*.

Given a rejection function R, if $0 \in R(\{0\} \cup F)$ then we interpret this as meaning that the agent finds at least one gamble f in F desirable, or in other words, prefers f over 0, and thus F should be a desirable gamble set. Then $K \coloneqq \{F \in \mathcal{Q} : 0 \in R(\{0\} \cup F)\}$ is the set of desirable gamble sets corresponding to R. Conversely, given a set of desirable gamble sets K, the corresponding rejection function R is given by $R(F) \coloneqq \{f \in F : F - \{f\} \in K\}$, for every F in $\mathcal{Q}$. Under conditions implied by coherence, both operations commute; see Ref. [3].

Definition 2 (Coherent sets of desirable gamble sets [3, Definition 4]). *A set of desirable gamble sets $K \subseteq \mathcal{Q}$ is called* coherent *if:*
K_0. $\emptyset \notin K$;
K_1. $F \in K \Rightarrow F \setminus \{0\} \in K$;
K_2. $\{f\} \in K$ *for all* f *in* $\mathcal{L}_{>0}$;
K_3. *if* $F, G \in K$ *and* $(\lambda^{f,g}, \mu^{f,g}) > 0$ *for every pair* (f, g) *in* $F \times G$, *then* $\{\lambda^{f,g} f + \mu^{f,g} g : f \in F, g \in G\} \in K$;[2]
K_4. *if* $F \in K$ *and* $F \subseteq G$, *then* $G \in K$.
We collect all the coherent sets of desirable gamble sets in the collection $\overline{\mathcal{K}}(\mathcal{X})$, or $\overline{\mathcal{K}}$ when it is clear from the context what the possibility space is.

We refer to Refs. [3,4] for a justification of the coherence axioms.

A set of desirable gamble sets K_1 is called *at least as informative as* a set of desirable gamble sets K_2 when $K_2 \subseteq K_1$, simply because an agent with K_1 finds more gamble sets desirable than an agent with K_2. The resulting partially ordered set $(\overline{\mathcal{K}}, \subseteq)$ is a complete meet-semilattice [3, Theorem 8]. This guarantees that if a partially specified set of desirable gamble sets $\mathcal{A} \subseteq \mathcal{Q}$ has a coherent superset, in which case $\mathcal{A}$ is called *consistent*, then it has a unique smallest (or least informative) coherent superset $\mathrm{cl}_{\overline{\mathcal{K}}}(\mathcal{A}) \coloneqq \bigcap\{K \in \overline{\mathcal{K}} : \mathcal{A} \subseteq K\}$, called the *natural extension* of $\mathcal{A}$ [3, Theorem 9].

[2] We use $(\lambda^{f,g}, \mu^{f,g}) > 0$ to mean '$\lambda^{f,g} \geq 0$ and $\mu^{f,g} \geq 0$ but not $\lambda^{f,g} = \mu^{f,g} = 0$.'

The natural extension $\mathrm{cl}_{\overline{\mathcal{K}}}(\mathcal{A})$ of a consistent assessment $\mathcal{A}$ obtains a closed-form expression; however, we will focus on its more convenient formulation in terms of *representations*, which we will discuss in Sections 2.2 and 2.3.

2.2 Sets of desirable gambles

Given a coherent rejection function R, the gambles in $D_R \coloneqq \{f \in \mathcal{L} : 0 \in R(\{0, f\})\}$ are preferred in a *pairwise preference* over 0. Similarly, for a coherent set of desirable gamble sets K, the set $D_K \coloneqq \{f \in \mathcal{L} : \{f\} \in K\}$ consists of the gambles that are preferred over 0. If R and K contain the same information, the sets D_R and D_K coincide. We therefore call D_K the *binary part* of K.

The set D_K is more commonly known as a *set of desirable gambles*: it contains the gambles that the agent prefers over 0. Sets of desirable gambles were introduced by Seidenfeld et al. [11], and have been further developed by a range of authors, for instance in Refs. [22,23,20,10,2].

Definition 3 (Sets of desirable gambles). *A set of desirable gambles D on $\mathcal{X}$ is a subset of $\mathcal{L}(\mathcal{X})$.*

The idea is that the set of desirable gambles D contains all the gambles f that are preferred over 0. Sets of desirable gambles have the following rationality axioms.

Definition 4 (Coherent sets of desirable gambles). *A set of desirable gambles $D \subseteq \mathcal{L}$ is called* coherent *if:*
D_1. $0 \notin D$;
D_2. $\mathcal{L}_{>0} \subseteq D$;
D_3. *if* $f, g \in D$ *and* $(\lambda, \mu) > 0$, *then* $\lambda f + \mu g \in D$.
We collect all the coherent sets of desirable gambles in $\overline{\mathcal{D}}(\mathcal{X}) \subseteq \mathscr{P}(\mathcal{L}(\mathcal{X}))$,[3] *or* $\overline{\mathcal{D}}$ *if it is clear from the context what the possibility space is.*

Coherent sets of desirable gambles are convex cones – sets $A \subseteq \mathcal{L}$ for which $\mathrm{posi}(A) \coloneqq \{\sum_{k=1}^{n} \lambda_k f_k : n \in \mathbb{N}, \lambda_k \in \mathbb{R}_{>0}, f_k \in A\} \subseteq A$ – that do not contain 0 but contain all the positive gambles. Given a probability mass function (pmf) p, we let $D_p \coloneqq \{f \in \mathcal{L} : E_p(f) > 0 \text{ or } f > 0\}$ be the corresponding set of desirable gambles, which collects all the gambles whose p-expectation E_p is positive, and the set $\mathcal{L}_{>0}$. It is a matter of straightforward verification that D_p is a coherent set of desirable gambles. If $p(x) > 0$ for all x in $\mathcal{X}$ then D_p is an open half space.

A set of desirable gambles D_1 is called *at least as informative as* a set of desirable gambles D_2 when $D_2 \subseteq D_1$. The resulting partially ordered set $(\overline{\mathcal{D}}, \subseteq)$ is also a complete meet-semilattice, and if a partially specified set of desirable gambles $A \subseteq \mathcal{L}$ has a coherent superset, in which case A is called *consistent*, then it has a unique smallest (or least informative) coherent superset $\mathrm{cl}_{\overline{\mathcal{D}}}(A) \coloneqq \bigcap\{D \in \overline{\mathcal{D}} : A \subseteq D\}$, called the *natural extension* of A. The natural extension for sets of desirable gambles has a convenient closed-form expression.

[3] In this paper, we will denote the powerset of a given set A by $\mathscr{P}(A)$, and the set of all non-empty subsets of A by $\mathscr{P}^+(A) = \mathscr{P}(A) \setminus \{\emptyset\}$.

Theorem 1 ([20, Theorem 1]). *Consider any assessment $A \subseteq \mathcal{L}$. Then A is consistent if and only if $0 \notin \operatorname{posi}(\mathcal{L}_{>0} \cup A)$. If this is the case, then $\operatorname{cl}_{\overline{\mathcal{D}}}(A) = \operatorname{posi}(\mathcal{L}_{>0} \cup A)$.*

Maximal sets of desirable gambles. The maximal elements of the partially ordered set $(\overline{\mathcal{D}}, \subseteq)$ are called *maximal sets of desirable gambles*. We collect them in the set $\mathcal{M}$.

Maximal sets of desirable gambles have an easy characterisation: a coherent set of desirable gambles D is maximal if and only if $f \in D$ or $-f \in D$ for all gambles $f \neq 0$ [20, Proposition 2]. This implies that every element of $\mathcal{M}(\mathcal{X})$ is a superset of D_p, where p is a pmf. As such, elements M of $\mathcal{M}(\mathcal{X})$ can be seen as more informative than probabilities: if a non-zero gamble f's expectation $E_p(f)$ is zero, then M still contains either f or $-f$.

For any set of gambles $A \subseteq \mathcal{L}$, denote the set of its dominating maximal sets of desirable gambles by $m(A) \coloneqq \{M \in \mathcal{M} : A \subseteq M\}$. Given a coherent set of desirable gambles D, the set $m(D)$ can be viewed as a representation of D.

Theorem 2 ([20, Corollary 4]). *A set of desirable gambles A is consistent if and only if there is some maximal M such that $A \subseteq M$. Moreover, its natural extension can be obtained by $\operatorname{cl}_{\overline{\mathcal{D}}}(A) = \bigcap m(A)$.*

In particular, Theorem 2 implies that any coherent set of desirable gambles D can be represented by the maximal sets of desirable gambles above it.

2.3 Representation

Given a coherent set of desirable gambles D, there may be multiple coherent sets of desirable gamble sets K such that $D_K = D$, but the smallest – or least informative – of them is $K_D \coloneqq \{F \in \mathcal{Q} : F \cap D \neq \emptyset\}$ [16, Proposition 5]. Because K_D is completely determined by D, and hence by *binary comparisons*, we call any coherent K_D a *binary* set of desirable gamble sets. Binary sets of desirable gambles are useful because they allow for a representation of any coherent set of desirable gamble sets.

Theorem 3 ([4, Theorem 9]). *For any sets of desirable gamble sets K, the following two expressions are equivalent:*

(i) K is coherent;

(ii) there is a non-empty set $\mathcal{D} \subseteq \overline{\mathcal{D}}$ of coherent sets of desirable gambles such that $K = \bigcap\{K_D : D \in \mathcal{D}\}$. If this is the case, then K is said to be represented *by $\mathcal{D}$, and $\mathcal{D}$ is a* representation *of K.*

Moreover, K's (unique) largest representing set is $\mathcal{D}_K \coloneqq \{D \in \overline{\mathcal{D}} : K \subseteq K_D\}$.

De Cooman and De Bock showed us in private communication that Theorem 3 implies the following convenient expression for the natural extension.

Theorem 4 (Due to De Cooman & De Bock). *An assessment $\mathcal{A} \subseteq \mathcal{Q}$ is consistent if and only if there is a D in $\overline{\mathcal{D}}$ such that $\mathcal{A} \subseteq K_D$. Moreover, if this is the case, then $\operatorname{cl}_{\overline{\mathcal{K}}}(\mathcal{A}) = \bigcap\{K_D : D \in \mathcal{D}_{\mathcal{A}}\}$.*

3 Three concepts of independence

In this section, we compare three notions of independence for sets of desirable gamble sets: epistemic independence, representation independence and S-independence. Each of them captures a different intuitive concept of independence. The most widely used one, epistemic independence, has been introduced in Ref. [22, Sections 9.1 and 9.2] for lower previsions and further developed for sets of desirable gambles in Ref [19] and sets of desirable gamble sets in Ref. [15]. S-independence has been introduced in Ref. [13] and studied in more detail for choice functions in Ref. [5]. We introduce a new notion in this context – which we will call representation independence – based on epistemic and strong products for sets of desirable gambles; see Ref. [19].

To prioritise conceptual clarity and minimise technicalities, we have decided to focus on independence between two uncertain variables X and Y, with finite possibility spaces $\mathcal{X}$ and $\mathcal{Y}$, respectively.

3.1 Marginalisation and conditioning

In this section, we will discuss sets of desirable gamble sets that represent beliefs about X and Y *jointly*, necessary to our discussion of independence next. The discussion and results in Section 2 above hold for the possibility space $\mathcal{X} \times \mathcal{Y}$ without restriction.

For any gamble f in the $|\mathcal{X}||\mathcal{Y}|$-dimensional linear space $\mathcal{L}(\mathcal{X} \times \mathcal{Y})$ and any $x \in \mathcal{X}$, the partial map $f(x, \cdot)$ is a gamble on $\mathcal{Y}$; and similarly, for any $y \in \mathcal{Y}$, the partial map $f(\cdot, y)$ is a gamble on $\mathcal{X}$. In this section, the roles of X and Y are symmetric; to avoid repetition we decide to focus on one of them, but *swapping X for Y* would yield analogous statements or results. It will be convenient to relate a gamble f on $\mathcal{X}$ to a gamble on $\mathcal{X} \times \mathcal{Y}$.

Definition 5 (Cylindrical extension). *Given any gamble f on $\mathcal{X}$, we let its* cylindrical extension *f^Y to $\mathcal{X} \times \mathcal{Y}$ be defined by $f^Y(x, y) := f(x)$ for all x in $\mathcal{X}$ and y in $\mathcal{Y}$. Similarly, given any set of gambles $F \subseteq \mathcal{L}(\mathcal{X})$, we let its* cylindrical extension *$F^Y \subseteq \mathcal{L}(\mathcal{X} \times \mathcal{Y})$ be defined as $F^Y := \{f^Y : f \in F\}$.*

Formally, f is a gamble on $\mathcal{X}$, while f^Y is a gamble on $\mathcal{X} \times \mathcal{Y}$, but since it is constant on $\mathcal{Y}$, its behavioural content is contained in $\mathcal{X}$ – just like for f. We will therefore, from now on, *make no notational distinction between a gamble and its cylindrical extension.* This notational trick allows us to regard f as an element of $\mathcal{L}(\mathcal{X} \times \mathcal{Y})$, and $\mathcal{L}(\mathcal{X})$ as a subset of $\mathcal{L}(\mathcal{X} \times \mathcal{Y})$ – even though their dimensions might differ – and similarly for $\mathcal{Q}(\mathcal{X}) \subseteq \mathcal{Q}(\mathcal{X} \times \mathcal{Y})$.

Marginalisation. Suppose that we have a coherent belief model D or K about (X, Y) jointly and we are interested in marginalising to X, or extracting the information about X solely. For a set of desirable gambles D, we collect the gambles that belong to D, but only depend on $\mathcal{X}$ in its X-marginal $\operatorname{marg}_X D$.

Definition 6 ([16]). *For any set of desirable gambles $D \subseteq \mathcal{L}(\mathcal{X} \times \mathcal{Y})$, its X-marginal $\operatorname{marg}_X D \subseteq \mathcal{L}(\mathcal{X})$ is defined as $\operatorname{marg}_X D \coloneqq D \cap \mathcal{L}(\mathcal{X})$.*

Similarly, for a set of desirable gamble sets K, we collect the gamble sets that belong to K, but only depend on $\mathcal{X}$, in its X-marginal $\operatorname{Marg}_X K$.

Definition 7 ([16]). *For any set of desirable gamble set $K \subseteq \mathcal{Q}(\mathcal{X} \times \mathcal{Y})$, its X-marginal $\operatorname{Marg}_X K \subseteq \mathcal{Q}(\mathcal{X})$ is defined as $\operatorname{Marg}_X K \coloneqq K \cap \mathcal{Q}(\mathcal{X})$.*

Definition 7 generalises Definition 6, in the sense that for all K' in $\overline{\mathcal{K}}$ and D' in $\overline{\mathcal{D}}$, we have [16, Proposition 10]

$$\operatorname{Marg}_X K_{D'} = K_{\operatorname{marg}_X D'} \text{ and } \operatorname{marg}_X D_{K'} = D_{\operatorname{Marg}_X K'}. \tag{1}$$

Marginalisation preserves the order for both sets of desirable gambles and set of desirable gamble sets: if $D_1 \subseteq D_2$, then $\operatorname{marg}_X D_1 \subseteq \operatorname{marg}_X D_2$ and if $K_1 \subseteq K_2$, then $\operatorname{Marg}_X K_1 \subseteq \operatorname{Marg}_X K_2$. Likewise, marginalisation preserves coherence: if a set of desirable gambles D is coherent, then so is $\operatorname{marg}_X D$; and if a set of desirable gamble sets K is coherent, then so is $\operatorname{Marg}_X K$. Moreover, using the lifted version $\operatorname{marg}_X \mathcal{D} \coloneqq \{\operatorname{marg}_X D : D \in \mathcal{D}\}$, if a coherent set of desirable gamble sets K is represented by $\mathcal{D}$, then $\operatorname{Marg}_X K$ is represented by $\operatorname{marg}_X \mathcal{D}$; see Ref. [15, Proposition 8].

Conditioning. Assume that we have a model D or K representing the agent's beliefs about the uncertain variable (X, Y) jointly. Suppose we receive new information that X takes values in some event $E \in \mathscr{P}^+(\mathcal{X})$. We aim to update our model to reflect this information, which is to *condition* on the event E.

We define the multiplication of a gamble f on $E \times \mathcal{Y}$ with the indicator $\mathbb{I}_E$ of E to be a gamble $\mathbb{I}_E f \in \mathcal{L}(\mathcal{X} \times \mathcal{Y})$ given by, for all $(x, y) \in \mathcal{X} \times \mathcal{Y}$:

$$\mathbb{I}_E f(x, y) \coloneqq \begin{cases} f(x, y), & \text{if } x \in E; \\ 0, & \text{otherwise.} \end{cases}$$

It is a *called-off* version of f: the risky transaction occurs only if E happens – if not, the gamble gets called off. In turn, the multiplication $\mathbb{I}_E$ with a gamble set A on $E \times \mathcal{Y}$ is $\mathbb{I}_E A = \{\mathbb{I}_E f : f \in A\}$.

Conditioning D on E yields the conditional set of desirable gambles

$$D \rfloor E \coloneqq \{f \in \mathcal{L}(E \times \mathcal{Y}) : \mathbb{I}_E f \in D\}$$

and if D is coherent, then so is $D \rfloor E$. We refer to [22, Appendix F] and [20, Section 3.2] for more information.

Similarly, conditioning K on E yields the conditional set of desirable gamble sets

$$K \rfloor E \coloneqq \{A \in \mathcal{Q}(E \times \mathcal{Y}) : \mathbb{I}_E A \in K\}$$

and if K is coherent, then so is $K \rfloor E$. Moreover, if $\mathcal{D}$ is some representation of K, then $K \rfloor E$ is represented by $\mathcal{D} \rfloor E = \{D \rfloor E : D \in \mathcal{D}\}$ [15, Proposition 7].

3.2 Independence concepts

In this section we provide an overview of three concepts of independence. In order to streamline the discussion, let K denote a coherent set of desirable gamble sets on the joint (finite) possibility space $\mathcal{X} \times \mathcal{Y}$.

Epistemic independence The idea of epistemic irrelevance from X to Y is:

> Learning that a non-trivial event $E_X \subseteq \mathcal{X}$ obtains does not affect the beliefs about Y.

The set K is said in Ref. [15, Section 6] to satisfy *epistemic irrelevance from X to Y* when conditioning on any non-trivial event $E_X \subseteq \mathcal{X}$ leaves the marginal beliefs about Y unchanged. Formally, for all E_X in $\mathscr{P}^+(\mathcal{X})$, $\mathrm{Marg}_Y(K\rfloor E_X) = \mathrm{Marg}_Y K$. Epistemic irrelevance is asymmetric, and to obtain its symmetric counterpart – called *epistemic independence* – one must require it in both directions. For K, this requirement is

$$\mathrm{Marg}_Y(K\rfloor E_X) = \mathrm{Marg}_Y K \text{ and } \mathrm{Marg}_X(K\rfloor E_Y) = \mathrm{Marg}_X K \tag{2}$$

for all E_X in $\mathscr{P}^+(\mathcal{X})$ and E_Y in $\mathscr{P}^+(\mathcal{Y})$. If Equation (2) holds for all such events, we say that K satisfies *epistemic independence*.

The K on $\mathcal{X} \times \mathcal{Y}$ is called an *epistemic product* of coherent K_X on $\mathcal{X}$ and K_Y on $\mathcal{Y}$ if it satisfies epistemic independence and has K_X and K_Y as its marginals. It is established in Ref. [15, Theorem 15] that the smallest epistemic product of K_X and K_Y exists, and we denote it by $K_1 \otimes K_2$.

The requirement of epistemic irrelevance can also be applied on sets of desirable gambles. A coherent set of desirable gambles D on $\mathcal{X} \times \mathcal{Y}$ is said in Ref. [19] to satisfy *epistemic irrelevance from X to Y* if $\mathrm{marg}_Y(D\rfloor E_X) = \mathrm{marg}_Y D$ for all E_X in $\mathscr{P}^+(\mathcal{X})$. This notion is again asymmetric, and to obtain a symmetric counterpart – called *epistemic independence* – one must require it in both directions. Similarly as for sets of desirable gamble sets, given two coherent sets of desirable gambles D_X on $\mathcal{X}$ and D_Y on $\mathcal{Y}$, a set D on $\mathcal{X} \times \mathcal{Y}$ is called an *epistemic product* of D_X and D_Y if it satisfies epistemic independence and has D_X and D_Y as its marginals. It is established in Ref. [19, Theorem 19] that the smallest epistemic product of D_X and D_Y exists, and we denote it by $D_X \otimes D_Y$.

Fortunately, for pmfs, the epistemic product for sets of desirable gambles coincides with the standard independence definition, in the following sense.

Proposition 1. *Consider a pmf p on $\mathcal{X}$ and a pmf q on $\mathcal{Y}$ such that $p(x) > 0$ for all x in $\mathcal{X}$ and $q(y) > 0$ for all y in $\mathcal{X}$. Then $D_{pq} = D_p \otimes D_q$.*

Representation independence We introduce a new independence requirement, which we call *representation independence*:

> There is an *independent representation*, that is, a representation whose every member satisfies epistemic independence.

Accordingly, we define K to satisfy representation independence whenever the following requirement is met.

Definition 8 (Representation independence). *A coherent set of desirable gamble sets K on $\mathcal{X} \times \mathcal{Y}$ satisfies* representation independence *if it has a representation $\mathcal{D}$ such that every D in $\mathcal{D}$ satisfies epistemic independence.*

Given two coherent sets of desirable gamble sets K_X on $\mathcal{X}$ and K_Y on $\mathcal{Y}$, we call K on $\mathcal{X} \times \mathcal{Y}$ a *representation product* of K_X and K_Y if it satisfies representation independence and has K_X and K_Y as its marginals. We establish that the smallest representation product of K_X and K_Y – which we will call the *representation-independent natural extension of K_X and K_Y* – exists and has a convenient representation in terms of the epistemic independent natural extension of the elements in the representations of K_X and K_Y.

Theorem 5. *Consider any coherent marginal sets of desirable gamble sets $K_X \in \overline{\mathcal{K}}(\mathcal{X})$ and $K_Y \in \overline{\mathcal{K}}(\mathcal{Y})$. Then the representation-independent natural extension of K_X and K_Y exists and is given by*

$$K_X \boxdot K_Y \coloneqq \bigcap_{D_X \in \mathcal{D}_{K_X}, D_Y \in \mathcal{D}_{K_Y}} K_{D_X \otimes D_Y}, \tag{3}$$

which has $\{D_X \otimes D_Y : D_X \in \mathcal{D}_{K_X}, D_Y \in \mathcal{D}_{K_Y}\}$ as an independent representation.

The concept of representation independence for sets of desirable gamble sets is similar to the *strong product* of sets of desirable gambles [19, Theorem 2]. There, given coherent sets of desirable gambles D_X on $\mathcal{X}$ and D_Y on $\mathcal{Y}$, their strong product is defined as:

$$D_X \boxtimes D_Y \coloneqq \bigcap \{M_X \otimes M_Y : M_X \in m(D_X), M_Y \in m(D_Y)\}. \tag{4}$$

Similarly to the representation-independent product, the strong product is an intersection of epistemic products of the members of representations $m(D_X)$ and $m(D_Y)$ of D_X and D_Y, respectively. In the context of coherent sets of desirable gambles, [19, Proposition 27] establishes that $D_X \boxtimes D_Y$ is indeed an epistemic product of D_X and D_Y, but not necessarily the smallest one [19, Appendix A.3].

This tells us that there is a choice to be made in Equation (3) about the type of product of the D_X and D_Y. In our Definition 8 of representation independence, we have chosen to require *epistemic* independence, but the results in Ref. [19] allows us to use a *strong* product instead. We thus obtain a different independence notion, which we call *strong representation independence*.

Definition 9 (Strong representation independence). *A coherent set of desirable gamble sets K on $\mathcal{X} \times \mathcal{Y}$ satisfies* strong representation independence *if it has a representation $\mathcal{D}$ such that every D in $\mathcal{D}$ is a strong product.*

Since every strong product is an epistemic product, if K satisfies strong representation independence, then it also satisfies representation independence. Similar to what we have done above, we call K on $\mathcal{X} \times \mathcal{Y}$ a *strong representation product*

of the two coherent sets of desirable gamble sets K_X on $\mathcal{X}$ and K_Y on $\mathcal{Y}$, if it satisfies strong representation independence and has K_X and K_Y as its marginals. Here too, it turns out that the smallest representation product of K_X and K_Y – which we will call the *strong representation-independent natural extension of* K_X *and* K_Y – exists and has a convenient representation in terms of strong products of elements in the representations of K_X and K_Y.

Theorem 6. *Consider any coherent marginal sets of desirable gamble sets* $K_X \in \overline{\mathcal{K}}(\mathcal{X})$ *and* $K_Y \in \overline{\mathcal{K}}(\mathcal{Y})$*. Then the strong representation-independent natural extension of* K_X *and* K_Y *exists and is given by*

$$K_X \boxdot K_Y := \bigcap_{K_X \in \mathcal{D}_{K_X}, K_Y \in \mathcal{D}_{K_Y}} K_{D_X \boxtimes D_Y}, \tag{5}$$

which has $\{D_X \boxtimes D_Y : D_X \in \mathcal{D}_{K_X}, D_Y \in \mathcal{D}_{K_Y}\}$ *as an independent representation.*

Interestingly, epistemic independence is a weaker requirement than (stong) representation independence.

Proposition 2. *Consider any coherent set of desirable gamble sets* K *on* $\mathcal{X} \times \mathcal{Y}$*. If* K *satisfies representation independence or strong representation independence, then* K *satisfies epistemic independence.*

As a consequence, the (strong) representation product of two coherent sets of desirable gamble sets K_X and K_Y is an epistemic product of K_X and K_Y, but it is not necessarily the smallest one.

Corollary 1. *Consider any coherent sets of desirable gamble sets* $K_X \in \overline{\mathcal{K}}(\mathcal{X})$ *and* $K_Y \in \overline{\mathcal{K}}(\mathcal{Y})$*. Then* $K_X \otimes K_Y \subseteq K_X \boxdot K_Y \subseteq K_X \boxdot K_Y$*.*

S-independence The idea of *S-irrelevance* of X to Y is [5, Section 4.1]:

> It is unreasonable to *spend resources* in order to use the observed value of X to choose between gambles that only depend on Y.

When X is S-irrelevant to Y and Y is S-irrelevant to X, the two uncertain variables are called *S-independent*.

Observing the value of X is modelled by selecting a partition $\mathcal{P}$ of the possibility space $\mathcal{X}$ and determining which event $E \in \mathcal{P}$ occurs, so that $X \in E$. For each event E in the partition, the agent chooses the gamble f_E on $\mathcal{Y}$. The action of observing X and then selecting a gamble on Y in this manner corresponds to the composite gamble $\sum_{E \in \mathcal{P}} \mathbb{I}_E(X) f_E(Y)$.

If X is S-irrelevant to Y, the requirement is that the agent choosing between the gambles on Y should reject the option to pay any real $\epsilon > 0$ for the composite gamble: from the gamble set $\{f_E : E \in \mathcal{P}\} \cup \{\sum_{G \in \mathcal{P}} \mathbb{I}_G f_G - \epsilon\}$, the agent should reject the latter gamble. Translated to sets of desirable gamble sets, this requirement becomes as follows [5, Definition 9].

Definition 10 (S-independence). *A coherent set of desirable gamble sets K on $\mathcal{X} \times \mathcal{Y}$ satisfies* S-irrelevance from X to Y *if* $\{\sum_{G \in \mathcal{P} \setminus \{E\}} \mathbb{I}_G[f_E - f_G] + \epsilon : E \in \mathcal{P}\} \in K$*, for all partitions $\mathcal{P}$ of $\mathcal{X}$, all $f_E \in \mathcal{L}(\mathcal{Y})$ and all $\epsilon > 0$. K satisfies* S-independent *if it satisfies S-irrelevance both from X to Y and from Y to X.*

Using representations of K, we can simplify the requirement in Definition 10, which will be useful later on this paper, thereby generalising [5, Proposition 12].

Proposition 3. *Consider any coherent set of desirable gamble sets K on $\mathcal{X} \times \mathcal{Y}$, and any representation $\mathcal{D}$ of K. Then K satisfies S-irrelevance from X to Y if and only if for all D in $\mathcal{D}$, the binary K_D satisfies S-irrelevance from X to Y.*

The theory of S-independence was significantly developed in Ref. [5]. To fully appreciate and build upon this work, we must introduce two additional axioms for sets of desirable gamble sets, the first being *mixingness*. A coherent set of desirable gamble sets K satisfies mixingness if
K_5. If $G \in K$ and $F \subseteq G \subseteq \operatorname{posi} F$, then also $F \in K$.
The effect of adding this axiom, is that any mixing set of desirable gamble sets K has a representation $\mathcal{D}$ – see Theorem 3 – that consists of *mixing sets of desirable gambles*, which are coherent sets of desirable gambles D for which the complement is also a convex cone: $\operatorname{posi}(D^c) = D^c$.[4] All maximal sets of desirable gambles, as well as all D_p for pmfs p such that $p(x) > 0$ for every x in $\mathcal{X}$, are mixing.

The second additional axiom is *Archimedeanity*, for which we refer to Ref. [18] for a detailed description. What matters for our analysis is that, once this requirement is imposed on K, it can be represented by $\mathcal{D}$ containing only sets of desirable gambles of the form $D_{\underline{P}} := \{f \in \mathcal{L} : \underline{P}(f) > 0\}$, where $\underline{P}$ is a coherent lower prevision. A coherent lower prevision is a lower expectation functional; we refer to [22,9] for more information about lower previsions. This means that we can represent an Archimedean set of desirable gamble sets K by a set $\underline{\mathcal{P}}$ of coherent lower previsions, and its largest representation in terms of lower previsions is $\underline{\mathcal{P}}_K := \{\underline{P}$ coherent lower prevision $: K \subseteq K_{D_{\underline{P}}}\}$; see Ref. [18] for more details.

A *linear prevision*, or (precise) expectation operator, is a lower prevision that corresponds to a precise probability. Imposing both Archimedeanity and mixingness on K has the effect that the set $\underline{\mathcal{P}}_K$ consists of linear previsions only; see [5, Theorem 2]. Together with a technical condition called *credible indeterminacy*,[5] the following surprising result is established in Ref [5].

Theorem 7 ([5, Theorem 10]). *Consider any Archimedean set of desirable gamble sets K on $\mathcal{X} \times \mathcal{Y}$ whose X- and Y-marginals are credibly indeterminate. If K satisfies S-independence, then every $\underline{P}$ in $\underline{\mathcal{P}}_K$ is a linear prevision that moreover satisfies independence.*

[4] Such sets of desirable gambles have also been called *lexicographic* in Refs. [17,1].

[5] For convenience, we use a stronger version of credible indeterminacy than in Ref. [5], but we will give it the same name. This guarantees that under our more restrictive requirements, the results in Ref. [5] remain valid. We call a set of desirable gamble sets K_X on $\mathcal{X}$ *credibly indeterminate* if for every outcome x in $\mathcal{X}$ there is a real $\epsilon > 0$ such that $\{\mathbb{I}_{\{x\}} - \epsilon\} \in K$.

Theorem 7 and Proposition 1 imply that Archimedean and credibly indeterminate sets of desirable gamble sets that satisfy S-independence, also satisfy representation independence. We call K on $\mathcal{X} \times \mathcal{Y}$ an *S-independent product* of two coherent sets of desirable gamble sets K_X on $\mathcal{X}$ and K_Y on $\mathcal{Y}$ if it is Archimedean, satisfies S-independence and has K_X and K_Y as its marginals. It turns out that this independence concept, too, has a smallest S-independent product of K_X and K_Y, at least when K_X and K_Y are mixing, Archimedean and credibly indeterminate, which we call the *S-independent natural extension.*

Theorem 8. *Consider Archimedean, mixing and credibly indeterminate sets of desirable gamble sets $K_X \in \overline{\mathcal{K}}(\mathcal{X})$ and $K_Y \in \overline{\mathcal{K}}(\mathcal{Y})$. Then the S-independent natural extension of K_X and K_Y exists and is given by*

$$K_X \circledast K_Y \coloneqq \bigcap_{P_X \in \underline{\mathcal{P}}_{K_X}, P_Y \in \underline{\mathcal{P}}_{K_Y}} K_{D_{P_X P_Y}}, \tag{6}$$

which has $\{D_{P_X P_Y} : P_X \in \underline{\mathcal{P}}_{K_X}, P_Y \in \underline{\mathcal{P}}_{K_Y}\}$ as an (independent) representation.

It turns out that the new notion of representation independence is expressive enough to connect K to S-independence without the need to impose Archimedeanity, as is required in Theorem 7.

Proposition 4. *Consider any mixing set of desirable gamble sets K on $\mathcal{X} \times \mathcal{Y}$. If K satisfies (strong) representation independence, then K satisfies S-independence.*

4 Discussion

We have introduced a new notion of independence for sets of desirable gamble sets – representation independence – together with its variant, strong representation independence. For these notions, as well as for the existing notion of S-independence, we established independent natural extensions of marginal sets of desirable gamble sets, and compared them with the epistemic independent natural extension.

We verify that epistemic independence, representation independence, and S-independence all coincide with the standard notion of independence for probability measures, in the following sense. Let p and q be pmfs on $\mathcal{X}$ and $\mathcal{Y}$, respectively, such that $p(x) > 0$ and $q(y) > 0$ for all $x \in \mathcal{X}$ and $y \in \mathcal{Y}$, and define $K_p \coloneqq K_{D_p}$, and analogously for q and pq. Then the results in this paper guarantee that

$$K_p \otimes K_q = K_p \boxcircle K_q = K_p \circledast K_q = K_{pq}.$$

Indeed, Proposition 1 yields $K_p \otimes K_q = K_{pq}$. Since K_p is Archimedean, mixing, and credibly indeterminate with singleton representation $\underline{\mathcal{P}}_{K_p} = \{E_p\}$ (and similarly for q and pq), Theorem 8 implies $K_p \circledast K_q = K_{pq}$. As the S-independent product is a representation product, it is a superset of the smallest such product $K_p \boxcircle K_q$. Together with Proposition 2, this gives $K_{pq} = K_p \otimes K_q \subseteq K_p \boxcircle K_q \subseteq K_p \circledast K_q = K_{pq}$, and hence all three products coincide in this case.

Several directions for further research remain open. We have shown that strong independence for mixing sets of desirable gamble sets implies S-independence, but it is still an open question whether this result can be strengthened to an equivalence, and under which conditions this would be possible. Additionally, the relation with mixingness opens up more questions, such as whether there are products that preserve mixingness. We also wonder to which extent the Archimedeanity condition is necessary in Ref. [5], and whether there are ways to obtain an S-independent product without relying on Archimedeanity.

One of the motivations for this work is to obtain a suitable notion of independence for use in Bayesian networks with sets of desirable gambles as local models. For this purpose, the associated independent product should ideally be associative. Whether any of the proposed products satisfies associativity is still unknown. Furthermore, for iterative applications of independent products, we must either find a canonical representation of the product, or allow for the representations of the marginals to be arbitrary, neither of which is currently established for any of the products.

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Appendix

Proof (Proof of Proposition 1). We will prove (i) that $D_{pq} \subseteq D_p \otimes D_q$ and (ii) that $D_{pq} \supseteq D_p \otimes D_q$.

For (i), we use [19, Theorem 21] to infer that the associated lower prevision $\underline{P}_{pq}$ of $D_p \otimes D_q$ coincides with the epistemic product of the associated lower previsions $\underline{P}_p$ and $\underline{P}_q$ of D_p and D_q. The lower previsions $\underline{P}_p$ and $\underline{P}_q$ are the expectation operators E_p and E_q, respectively, so we may invoke the fact [7, Proposition 25] that their epistemic product is the expectation operator E_{pq} associated with the (independent) product pq. This implies that $\underline{P}_{pq}$ coincides with the linear prevision E_{pq} associated with D_{pq}. Since $D_{pq} = \{f \in \mathcal{L} : E_{pq}(f) > 0 \text{ or } f > 0\}$ is the smallest coherent set of desirable gambles that induces the linear prevision E_{pq}, we infer that $D_{pq} \subseteq D_p \otimes D_q$.

For (ii), we will establish that D_{pq} is an epistemic product of D_p and D_q. Since $D_p \otimes D_q$ is the smallest such product, it will then follow that $D_{pq} \supseteq D_p \otimes D_q$. To show that D_{pq} is an epistemic product of D_p and D_q, we will verify that it has the correct marginals and satisfies epistemic independence. First, we verify that D_{pq} has the correct marginals. For any f in $\mathcal{L}(\mathcal{X})$, we have that f belongs to D_{pq} if and only if $\sum_{x\in\mathcal{X},y\in\mathcal{Y}} p(x)q(y)f(x) = \sum_{x\in\mathcal{X}} p(x)f(x) > 0$ or $f > 0$, which holds if and only if f belongs to D_p, so $\operatorname{marg}_X D_{pq} = D_p$. A similar argument shows that $\operatorname{marg}_Y D_{pq} = D_q$.

Finally, we verify that D_{pq} satisfies epistemic independence. For any E in $\mathscr{P}^+(\mathcal{X})$ and any f in $\mathcal{L}(\mathcal{Y})$, infer that

$$\begin{aligned} f \in \operatorname{marg}_Y(D_{pq}\rfloor E) &\Leftrightarrow f \in D_{pq}\rfloor E \Leftrightarrow \mathbb{I}_E f \in D_{pq} \\ &\Leftrightarrow \sum_{x\in E, y\in\mathcal{Y}} p(x)q(y)f(y) = E_q(f) \underbrace{\sum_{x\in E} p(x)}_{>0} > 0 \text{ or } \mathbb{I}_E f > 0 \\ &\Leftrightarrow E_q(f) > 0 \text{ or } f > 0 \Leftrightarrow f \in D_q = \operatorname{marg}_Y(D_{pq}), \end{aligned}$$

where the first equivalence follows from marginalisation because f's domain is $\mathcal{Y}$, and the third equivalence holds because $\mathbb{I}_E f > 0 \Leftrightarrow f > 0$ since $\mathbb{I}_E \in \mathcal{L}(\mathcal{X})$ and $f \in \mathcal{L}(\mathcal{Y})$. This implies that $\operatorname{marg}_Y(D_{pq}\rfloor E) = \operatorname{marg}_Y D_{pq}$ for any E in $\mathscr{P}^+(\mathcal{X})$, and a similar argument tells us that $\operatorname{marg}_X(D_{pq}\rfloor E) = \operatorname{marg}_X D_{pq}$ for any E in $\mathscr{P}^+(\mathcal{Y})$, so that, indeed, D_{pq} satisfies epistemic independence. □

Lemma 1. *Consider any non-empty $\mathcal{D}_1, \mathcal{D}_2 \subseteq \overline{\mathcal{D}}$. If $\mathcal{D}_2 \subseteq \mathcal{D}_1$ and $(\forall D_1 \in \mathcal{D}_1)(\exists D_2 \in \mathcal{D}_2) D_2 \subseteq D_1$, then $\mathcal{D}_1$ and $\mathcal{D}_2$ represent the same set of desirable gamble sets.*

Proof (Proof of Lemma 1). We will prove that

$$K_1 := \bigcap\{K_{D_1} : D_1 \in \mathcal{D}_1\} = \bigcap\{K_{D_2} : D_2 \in \mathcal{D}_2\} =: K_2.$$

From $\mathcal{D}_2 \subseteq \mathcal{D}_1$ we infer that $K_1 \subseteq K_2$ because the latter is an intersection over a smaller set, so it suffices to prove that also $K_2 \subseteq K_1$. To this end, consider any

F in K_2, so $F \in K_{D_2}$ and hence $F \cap D_2 \neq \emptyset$ for all D_2 in $\mathcal{D}_2$. Consider any D_1 in $\mathcal{D}_1$. By assumption, there is a D_2 in $\mathcal{D}_2$ such that $D_2 \subseteq D_1$, and therefore $F \cap D_1 \supseteq F \cap D_2 \neq \emptyset$, which implies that F belongs to K_{D_1}. This holds for any choice of D_1 in $\mathcal{D}_1$, so $F \in \bigcap\{K_{D_1} : D_1 \in \mathcal{D}_1\} = K_1$. Since the choice of F in K_2 was arbitrary, we infer that, indeed, $K_2 \subseteq K_1$. □

Proof (Proof of Theorem 5). Equation (3) is precisely what it means for the set $\{D_X \otimes D_Y : D_X \in \mathcal{D}_{K_X}, D_Y \in \mathcal{D}_{K_Y}\}$ to be a representation of $K_X \boxcircle K_Y$. Moreover, each element $D_X \otimes D_Y$ of it satisfies epistemic independence, so it is an independent representation. So it's left to establish the first statement. We will show that (i) $K_X \boxcircle K_Y$ is a representation product of K_X and K_Y, and (ii) that $K_X \boxcircle K_Y$ is the *smallest* such product.

To show (i) that $K_X \boxcircle K_Y$ is a representation product of K_X and K_Y, infer using Theorem 3 that it is coherent because the representation $\{D_X \otimes D_Y : D_X \in \mathcal{D}_{K_X}, D_Y \in \mathcal{D}_{K_Y}\}$ is non-empty and every $D_X \otimes D_Y$ in it is coherent. Also, this representation of $K_X \boxcircle K_Y$ consists only of epistemic products $D_X \otimes D_Y$, so $K_X \boxcircle K_Y$ satisfies representation independence by definition. Therefore, to establish that $K_X \boxcircle K_Y$ is a representation product of K_X and K_Y, it is left to prove that $\mathrm{Marg}_X(K_X \boxcircle K_Y) = K_X$ and $\mathrm{Marg}_Y(K_X \boxcircle K_Y) = K_Y$. Infer that, indeed,

$$\begin{aligned}
\mathrm{Marg}_X(K_X \boxcircle K_Y) &= \mathrm{Marg}_X\Big(\bigcap_{D_X \in \mathcal{D}_{K_X}, D_Y \in \mathcal{D}_{K_Y}} K_{D_X \otimes D_Y}\Big) \\
&= \bigcap_{D_X \in \mathcal{D}_{K_X}, D_Y \in \mathcal{D}_{K_Y}} \mathrm{Marg}_X(K_{D_X \otimes D_Y}) \\
&= \bigcap_{D_X \in \mathcal{D}_{K_X}, D_Y \in \mathcal{D}_{K_Y}} K_{\mathrm{marg}_X(D_X \otimes D_Y)} \\
&= \bigcap_{D_X \in \mathcal{D}_{K_X}, D_Y \in \mathcal{D}_{K_Y}} K_{D_X} = K_X,
\end{aligned}$$

where the second equality follows because marginalisation to X is an intersection with $\mathcal{Q}(\mathcal{X})$, the third equality follows from Equation (1), and the final equality follows from the fact, established in Ref. [19, Theorem 19], that $D_X \otimes D_Y$ is an epistemic product of D_X and D_Y. The proof that $\mathrm{Marg}_Y(K_X \boxcircle K_Y) = K_Y$ follows using a completely analogous argument.

We now turn to (ii). We will show that any representation product $K^\star$ of K_X and K_Y includes $K_X \boxcircle K_Y$. Once this is established, it follows that $K_X \boxcircle K_Y$ is the smallest representation product of K_X and K_Y, and the proof is complete.

So consider any representation product $K^\star$ of K_X and K_Y. By the definition of a representation product, $K^\star$ has a representation $\mathcal{D} \subseteq \mathcal{D}_{K^\star}$ such that every D in $\mathcal{D}$ satisfies epistemic independence. Moreover, $K^\star$ has K_X and K_Y as its marginals, so $K_X \subseteq K^\star$ and $K_Y \subseteq K^\star$, and therefore $K_X \cup K_Y \subseteq K^\star$. But this implies that $\mathcal{D}_{K^\star} \subseteq \mathcal{D}_{K_X \cup K_Y}$, whence $\mathcal{D} \subseteq \mathcal{D}_{K_X \cup K_Y}$. Moreover, since every member of $\mathcal{D}$ satisfies epistemic independence, we infer that

$$\mathcal{D} \subseteq \mathcal{D}_1 \coloneqq \mathcal{D}_{K_X \cup K_Y} \cap \{D \in \overline{\mathcal{D}}(\mathcal{X} \times \mathcal{Y}) : D \text{ satisfies epistemic independence}\}.$$

If we can show that the set $\mathcal{D}_1$ represents $K_X \boxdot K_Y$, we are done: since $\mathcal{D}$ is a subset of $\mathcal{D}_1$, the set of desirable gamble sets $K^\star$ represented by $\mathcal{D}$ then is necessarily, indeed, a superset of $K_X \boxdot K_Y$.

Using the earlier established fact that $\mathcal{D}_2 \coloneqq \{D_X \otimes D_Y : D_X \in \mathcal{D}_{K_X}, D_Y \in \mathcal{D}_{K_Y}\}$ represents $K_X \boxdot K_Y$ and invoking Lemma 1, the proof is done once we show that (a) $\mathcal{D}_2 \subseteq \mathcal{D}_1$ and (b) $(\forall D_1 \in \mathcal{D}_1)(\exists D_2 \in \mathcal{D}_2) D_2 \subseteq D_1$.

For (a), consider any D in $\mathcal{D}_2$. Then $D = D_X \otimes D_Y$ for some $D_X \in \mathcal{D}_{K_X}$ and $D_Y \in \mathcal{D}_{K_Y}$, so it satisfies epistemic independence. Moreover, D marginalises to D_X, whence $D \supseteq D_X$, so we infer that $K_X \subseteq K_{D_X} \subseteq K_D$. Similarly, we also find that $K_Y \subseteq K_D$. This implies that $D \in \mathcal{D}_1$, and therefore, indeed, $\mathcal{D}_2 \subseteq \mathcal{D}_1$.

For (b), consider any D in $\mathcal{D}_1$. Then D satisfies epistemic independence and $K_X \cup K_Y \subseteq K_D$. Let $D_X \coloneqq \operatorname{marg}_X D$ and $D_Y \coloneqq \operatorname{marg}_Y D$. Since $D_X \otimes D_Y$ is the smallest independent product of D_X and D_Y, we have $D \supseteq D_X \otimes D_Y$. Moreover, from $K_X \cup K_Y \subseteq K_D$ we infer that $K_X = \operatorname{Marg}_X(K_X \cup K_Y) \subseteq \operatorname{Marg}_X K_D = K_{D_X}$, and, using a similar argument, that $K_Y \subseteq K_{D_Y}$, so $D_X \in \mathcal{D}_{K_X}$ and $D_Y \in \mathcal{D}_{K_Y}$. So we have, indeed, found an element $D_X \otimes D_Y$ of $\mathcal{D}_2$ that is contained in D. □

Proof (Proof of Theorem 6). This proof is very similar to the one of Theorem 5. Since some details differ, we provide the full proof here, for completeness.

Equation (5) is precisely what it means for the set $\{D_X \boxtimes D_Y : D_X \in \mathcal{D}_{K_X}, D_Y \in \mathcal{D}_{K_Y}\}$ to be a representation of $K_X \boxdot K_Y$. Moreover, each element $D_X \boxtimes D_Y$ of it is a strong product, so it is an independent representation. To finish the proof, we establish the first statement. We will show that (i) $K_X \boxdot K_Y$ is a strong representation product of K_X and K_Y, and (ii) that $K_X \boxdot K_Y$ is the *smallest* such product.

To show (i) that $K_X \boxdot K_Y$ is a strong representation product of K_X and K_Y, infer using Theorem 3 that it is coherent because the representation $\{D_X \boxtimes D_Y : D_X \in \mathcal{D}_{K_X}, D_Y \in \mathcal{D}_{K_Y}\}$ is non-empty and every $D_X \boxtimes D_Y$ in it is coherent. Also, this representation of $K_X \boxdot K_Y$ consists only of strong products $D_X \boxtimes D_Y$, so $K_X \boxdot K_Y$ satisfies strong representation independence by definition. Therefore, to establish that $K_X \boxdot K_Y$ is a strong representation product of K_X and K_Y, it is left to prove that $\operatorname{Marg}_X(K_X \boxdot K_Y) = K_X$ and $\operatorname{Marg}_Y(K_X \boxdot K_Y) = K_Y$. Infer that, indeed,

$$\begin{aligned}
\operatorname{Marg}_X(K_X \boxdot K_Y) &= \operatorname{Marg}_X\Big(\bigcap_{D_X \in \mathcal{D}_{K_X}, D_Y \in \mathcal{D}_{K_Y}} K_{D_X \boxtimes D_Y}\Big) \\
&= \bigcap_{D_X \in \mathcal{D}_{K_X}, D_Y \in \mathcal{D}_{K_Y}} \operatorname{Marg}_X(K_{D_X \boxtimes D_Y}) \\
&= \bigcap_{D_X \in \mathcal{D}_{K_X}, D_Y \in \mathcal{D}_{K_Y}} K_{\operatorname{marg}_X(D_X \boxtimes D_Y)} \\
&= \bigcap_{D_X \in \mathcal{D}_{K_X}, D_Y \in \mathcal{D}_{K_Y}} K_{D_X} = K_X
\end{aligned}$$

where the second equality follows because marginalisation to X is an intersection with $\mathcal{Q}(\mathcal{X})$, the third equality follows from Equation (1), and the final equality

follows from the fact, established in Ref. [19, Proposition 27], that $D_X \boxtimes D_Y$ is an epistemic product of D_X and D_Y. The proof that $\mathrm{Marg}_Y(K_X \boxdot K_Y) = K_Y$ follows using a completely analogous argument.

We now turn to (ii). We will show that any strong representation product $K^\star$ of K_X and K_Y includes $K_X \boxdot K_Y$. Once this is established, it follows that $K_X \boxdot K_Y$ is the smallest strong representation product of K_X and K_Y, and the proof is complete.

So consider any strong representation product $K^\star$ of K_X and K_Y. By the definition of a strong representation product, $K^\star$ has a representation $\mathcal{D} \subseteq \mathcal{D}_{K^\star}$ such that every D in $\mathcal{D}$ is a strong product of some $D_X \in \overline{\mathcal{D}}(\mathcal{X})$ and $D_Y \in \overline{\mathcal{D}}(\mathcal{Y})$. Moreover, $K^\star$ has K_X and K_Y as its marginals, so $K_X \subseteq K^\star$ and $K_Y \subseteq K^\star$, and therefore $K_X \cup K_Y \subseteq K^\star$. But this implies that $\mathcal{D}_{K^\star} \subseteq \mathcal{D}_{K_X \cup K_Y}$, whence $\mathcal{D} \subseteq \mathcal{D}_{K_X \cup K_Y}$. Moreover, since every member of $\mathcal{D}$ is a strong product, we infer that

$$\mathcal{D} \subseteq \mathcal{D}_1 \coloneqq \mathcal{D}_{K_X \cup K_Y} \cap \{D \in \overline{\mathcal{D}}(\mathcal{X} \times \mathcal{Y}) : D \text{ is a strong product}\}.$$

We finish the proof by showing that $\mathcal{D}_1$ coincides with $\mathcal{D}_2 \coloneqq \{D_X \boxtimes D_Y : D_X \in \mathcal{D}_{K_X}, D_Y \in \mathcal{D}_{K_Y}\}$. Indeed, using the earlier established fact that $\mathcal{D}_2$ represents $K_X \boxdot K_Y$, the set of desirable gamble sets $K^\star$ represented by $\mathcal{D}$ then is necessarily, a superset of $K_X \boxdot K_Y$.

To show that $\mathcal{D}_1 \subseteq \mathcal{D}_2$, consider any D in $\mathcal{D}_1$. Then $D = D_X \boxtimes D_Y$ for some $D_X \in \overline{\mathcal{D}}(\mathcal{X})$ and $D_Y \in \overline{\mathcal{D}}(\mathcal{Y})$. But then D marginalises to D_X, so $K_X \cup K_Y \subseteq K_D$ implies that $K_X = \mathrm{Marg}_X(K_X \cup K_Y) \subseteq \mathrm{Marg}_X K_D = K_{D_X}$ and hence $D_X \in \mathcal{D}_{K_X}$. A similar argument allows us to infer that D_Y belongs to $\mathcal{D}_{K_Y}$. This implies that $D \in \mathcal{D}_2$, whence, indeed, $\mathcal{D}_1 \subseteq \mathcal{D}_2$.

Finally, to show that $\mathcal{D}_2 \subseteq \mathcal{D}_1$, consider any D in $\mathcal{D}_2$. Then $D = D_X \boxtimes D_Y$ for some $D_X \in \mathcal{D}_{K_X}$ and $D_Y \in \mathcal{D}_{K_Y}$, so D is a strong product. Moreover, D has D_X and D_Y as marginals, and therefore $K_X \cup K_Y \subseteq K_D$, so D belongs to $\mathcal{D}_1$, whence, indeed, $\mathcal{D}_2 \subseteq \mathcal{D}_1$. □

Proof (Proof of Proposition 2). If K satisfies strong representation independence, then it satisfies representation independence because a strong product of two coherent sets of desirable gambles is also an epistemic product, so it suffices to show that if K satisfies representation independence, then it also satisfies epistemic independence. So suppose that K satisfies representation independence, and we will show that then K satisfies epistemic independence. By symmetry, it suffices to show that $\mathrm{Marg}_Y(K \rfloor E) = \mathrm{Marg}_Y K$ for all E in $\mathscr{P}^+(\mathcal{X})$; the other equality where the roles of X and Y are swapped, follows using a completely analogous argument.

Since K satisfies representation independence, it has a representation $\mathcal{D}$ consisting only of sets of desirable gambles that satisfy epistemic independence. Consider any such set of desirable gambles D in $\mathcal{D}$, any gamble set F in $\mathcal{Q}(\mathcal{Y})$, and any conditioning event E in $\mathscr{P}^+(\mathcal{X})$, and infer the following equivalences.

$$\begin{aligned}\mathbb{I}_E F \in K_D \Leftrightarrow \mathbb{I}_E F \cap D \neq \emptyset &\Leftrightarrow F \cap D \rfloor E \neq \emptyset \\ &\Leftrightarrow F \cap \mathrm{marg}_Y(D \rfloor E) \neq \emptyset\end{aligned}$$

$$\Leftrightarrow F \cap \operatorname{marg}_Y D \neq \emptyset \Leftrightarrow F \cap D \neq \emptyset \Leftrightarrow F \in K_D,$$

where we used the definition of conditioning sets of desirable gambles in the second equivalence, the fact that F is a gamble set on $\mathcal{Y}$ in the third and fifth equivalence, and the fact that D is an epistemic product in the fourth equivalence. Since this equivalence holds for every D in $\mathcal{D}$, we infer that $\mathbb{I}_E F \in K$ if and only if $F \in K$, which implies that, indeed, $\operatorname{Marg}_Y(K \rfloor E_X) = \operatorname{Marg}_Y K$. □

Proof (Proof of Corollary 1). Proposition 2 implies that the representation independent $K_X \boxdot K_Y$ is an epistemic product, so it contains the smallest epistemic product, and therefore $K_X \otimes K_Y \subseteq K_X \boxdot K_Y$. Moreover, since $D_X \boxtimes D_Y$ is also an epistemic product, it includes the smallest epistemic product $D_X \otimes D_Y$, and thus, indeed, $K_X \boxdot K_Y = \bigcap\{K_{D_X \otimes D_Y} : D_X \in \mathcal{D}_{K_X}, D_Y \in \mathcal{D}_{K_{D_Y}}\} = \bigcap\{K_{D_X \boxtimes D_Y} : D_X \in \mathcal{D}_{K_X}, D_Y \in \mathcal{D}_{K_{D_Y}}\} \subseteq K_X \boxdot K_Y$. □

Proof (Proof of Proposition 3). Since $\mathcal{D}$ represents K, we have that $K = \bigcap\{K_D : D \in \mathcal{D}\}$. This implies that for all partitions $\mathcal{P}$ of $\mathcal{X}$, all $f_E \in \mathcal{L}(\mathcal{Y})$ and all $\epsilon > 0$, the set

$$\Big\{ \sum_{G \in \mathcal{P} \setminus \{E\}} \mathbb{I}_G[f_E - f_G] + \epsilon : E \in \mathcal{P} \Big\}$$

belongs to K if and only if it belongs to K_D for every D in $\mathcal{D}$. Definition 10 then implies that, indeed, K satisfies S-irrelevance from X to Y if and only if, for every D in $\mathcal{D}$, the binary K_D satisfies S-irrelevance from X to Y. □

Lemma 2. *Consider any mixing, Archimedean and credibly indeterminate set of desirable gamble sets K on $\mathcal{X}$. Then for every P in its representation $\underline{\mathcal{P}}_K$ and every x in $\mathcal{X}$ we have $P(\mathbb{I}_{\{x\}}) > 0$.*

Proof (Proof of Lemma 2). Because K is mixing and Archimedean, its largest representation $\underline{\mathcal{P}}_K$ consists of linear previsions. Consider any P in $\underline{\mathcal{P}}_K$ and any x in $\mathcal{X}$. Taking into account the credible indeterminacy of K, there is some $\epsilon > 0$ for which the singleton gamble set $\{\mathbb{I}_{\{x\}} - \epsilon\}$ belongs to K, and therefore $\mathbb{I}_{\{x\}} - \epsilon \in D_K$. This implies that $\mathbb{I}_{\{x\}} - \frac{\epsilon}{2} \in D_P$ which in turn implies that $P(\mathbb{I}_{\{x\}} - \frac{\epsilon}{2}) \geq 0$ and hence, indeed, $P(\mathbb{I}_{\{x\}}) > 0$. □

Proof (Proof of Theorem 8). Equation (6) is precisely what it means for the set $\{D_{P_X P_Y} : P_X \in \underline{\mathcal{P}}_{K_X}, P_Y \in \underline{\mathcal{P}}_{K_Y}\}$ to be a representation of $K_X \circledast K_Y$. Moreover, Lemma 2 tells us that for every x in $\mathcal{X}$ and y in $\mathcal{Y}$, and every P_X in $\underline{\mathcal{P}}_{K_X}$ and P_Y in $\underline{\mathcal{P}}_{K_Y}$, we have $P_X(\mathbb{I}_{\{x\}}) > 0$ and $P_Y(\mathbb{I}_{\{y\}}) > 0$, so that $P_X P_Y(\mathbb{I}_{\{(x,y)\}}) = P_X(\mathbb{I}_{\{x\}}) P_Y(\mathbb{I}_{\{y\}}) > 0$. Proposition 1 then implies that each element $D_{P_X P_Y} = D_{P_X} \otimes D_{P_Y}$ of $K_X \circledast K_Y$'s representation is an epistemic product. To finish the proof, we establish the first statement. We will show that (i) $K_X \circledast K_Y$ is an S-independent product of K_X and K_Y, and (ii) that $K_X \circledast K_Y$ is the *smallest* such product.

To show (i) that $K_X \circledast K_Y$ is an S-independent product of K_X and K_Y, infer using Theorem 3 that it is coherent because it has a non-empty representation, established above. Moreover, a very similar argument as in the proof of Theorem 5

shows that it has K_X and K_Y as its marginals. To show that it satisfies S-independence, we will show that it satisfies S-irrelevance from X to Y; a similar argument then allows us to infer that it also satisfies S-irrelevance from Y to X, and therefore S-independence. Taking into account Proposition 3, it suffices to show that

$$F := \Big\{ \sum_{G \in \mathcal{P} \setminus \{E\}} \mathbb{I}_G[f_E - f_G] + \epsilon : E \in \mathcal{P} \Big\} \in K_{D_{P_X P_Y}}$$

for all partitions $\mathcal{P}$ of $\mathcal{X}$, all $f_E \in \mathcal{L}(\mathcal{Y})$, all $\epsilon > 0$ all $P_X \in \underline{\mathcal{P}}_{K_X}$ and all $P_Y \in \underline{\mathcal{P}}_{K_Y}$. So consider any partition $\mathcal{P}$ of $\mathcal{X}$, f_E in $\mathcal{L}(\mathcal{Y})$, $\epsilon > 0$, P_X in $\underline{\mathcal{P}}_{K_X}$ and P_Y in $\underline{\mathcal{P}}_{K_Y}$. Then the $P_X P_Y$-expectation of $\sum_{G \in \mathcal{P} \setminus \{E\}} \mathbb{I}_G[f_E - f_G] + \epsilon$ is

$$\sum_{G \in \mathcal{P} \setminus \{E\}} P_X(\mathbb{I}_G)[P_Y(f_E - f_G)] + \epsilon.$$

Note that not for all E in $\mathcal{P}$, the expectation $\sum_{G \in \mathcal{P} \setminus \{E\}} P_X(\mathbb{I}_G)[P_Y(f_E - f_G)] < 0$. Indeed, if it was, then also its sum $\sum_{E \in \mathcal{P}} \sum_{G \in \mathcal{P} \setminus \{E\}} P_Y(f_E - f_G) = 0$ must be negative, which is a contradiction. So there is at least one E in $\mathcal{P}$ such that $\sum_{G \in \mathcal{P} \setminus \{E\}} P_X(\mathbb{I}_G)[P_Y(f_E - f_G)] \geq 0$, and therefore the $P_X P_Y$-expectation of $\sum_{G \in \mathcal{P} \setminus \{E\}} \mathbb{I}_G[f_E - f_G] + \epsilon$ is positive, which implies that F belongs to $K_{D_{P_X P_Y}}$, indeed. Finally, to establish that $K_X \circledast K_Y$ is Archimedean, simply observe that it has a representation of linear previsions, so it is Archimedean (even strongly Archimedean) by a Representation theorem for strongly Archimedean choice models [5, Theorem 1].

We now turn to (ii). We will show that any S-independent product $K^\star$ of K_X and K_Y includes $K_X \circledast K_Y$. Once this is established, it follows that $K_X \circledast K_Y$ is the smallest S-independent product of K_X and K_Y, and the proof is complete.

So consider any S-independent product $K^\star$ of K_X and K_Y. Since $K^\star$ is Archimedean, every P in $\underline{\mathcal{P}}_{K^\star}$ is a linear prevision that moreover satisfies independence, taking into account Theorem 7. Consider any P in $\underline{\mathcal{P}}_{K^\star}$, then $\operatorname{marg}_X P$ belongs to $\underline{\mathcal{P}}_{K_X}$ and $\operatorname{marg}_Y P$ belongs to $\underline{\mathcal{P}}_{K_Y}$, so $P = \operatorname{marg}_X P \operatorname{marg}_Y P$ and therefore P belongs to $\{P_X P_Y : P_X \in \underline{\mathcal{P}}_{K_X}, P_Y \in \underline{\mathcal{P}}_{K_Y}\}$. So we have established that $\underline{\mathcal{P}}_{K^\star} \subseteq \{P_X P_Y : P_X \in \underline{\mathcal{P}}_{K_X}, P_Y \in \underline{\mathcal{P}}_{K_Y}\}$, and therefore, indeed, $K_X \circledast K_Y = \bigcap_{P_X \in \underline{\mathcal{P}}_{K_X}, P_Y \in \underline{\mathcal{P}}_{K_Y}} K_{D_{P_X P_Y}} \subseteq \bigcap_{P \in \underline{\mathcal{P}}_{K^\star}} K_{D_P} = K^\star$. □

Lemma 3. *Consider any mixing sets of desirable gamble sets K. Then every element D in $\mathcal{D}_K$ is a superset of D_p for some pmf p.*

Proof. We will prove the contraposition. So consider any D in $\mathcal{D}_K$ that is not a superset of a D_p for some pmf p. Then D's associated coherent lower prevision $\underline{P}_D$, defined as

$$\underline{P}_D(f) := \sup\{\mu \in \mathbb{R} : f - \mu \in D\} \quad \text{for all } f \text{ in } \mathcal{L},$$

is not a linear prevision, so there is a gamble f whose upper prevision $\overline{P}_D(f) := -\underline{P}_D(-f)$ is strictly greater than $\underline{P}_D(f)$, so $\overline{P}_D(f) > \underline{P}_D(f)$. Let $\alpha := \frac{1}{2}(\overline{P}_D(f) +$

$\underline{P}_D(f))$ and $g := f - \alpha$. Since every coherent lower prevision and associated upper prevision is constant additive, we infer that

$$\underline{P}_D(g) = \underline{P}_D(f) - \frac{1}{2}(\overline{P}_D(f) + \underline{P}_D(f)) = -\frac{1}{2}(\overline{P}_D(f) - \underline{P}_D(f)) < 0$$

and

$$\overline{P}_D(g) = \overline{P}_D(f) - \frac{1}{2}(\overline{P}_D(f) + \underline{P}_D(f)) = \frac{1}{2}(\overline{P}_D(f) - \underline{P}_D(f)) > 0,$$

whence $\underline{P}_D(-g) = -\overline{P}_D(g) < 0$.

Consider any ϵ in the open interval $\big(0, -\max\{\underline{P}_D(g), \underline{P}_D(-g)\}\big)$, then, using constant additivity again, we infer that $\underline{P}_D(-g+\epsilon) < 0$ and $\underline{P}_D(g+\epsilon) < 0$. This implies that $-g+\epsilon$ and $g+\epsilon$ do not belong to D. By letting $F := \{g+\epsilon, -g+\epsilon\}$ and $G := \{\epsilon, g+\epsilon, -g+\epsilon\}$, we have that $G \in K$ by coherence and $F \subseteq G \subseteq \operatorname{posi} F$, but $F \notin K$ because $D \in \mathcal{D}_K$, which contradicts the requirement K_5 of mixingness. □

Proof (Proof of Proposition 4). By symmetry, it suffices to show that D satisfies S-irrelevance from X to Y. Taking Proposition 3 into account, it suffices to show, for all partitions $\mathcal{P}$ of $\mathcal{X}$, all $f_E \in \mathcal{L}(\mathcal{Y})$, all $\epsilon > 0$ and all D in a representation $\mathcal{D}$ of K, that the binary K_D contains the gamble set

$$F := \Big\{ \sum_{G \in \mathcal{P} \setminus \{E\}} \mathbb{I}_G[f_E - f_G] + \epsilon : E \in \mathcal{P} \Big\}.$$

Use Theorem 5 to infer that K has a representation $\mathcal{D}$ that consists of sets of desirable gambles that satisfy epistemic independence. Consider any D in $\mathcal{D}$. Since K is mixing, Lemma 3 then implies that $D \supseteq D_p$ for some pmf p on $\mathcal{X} \times \mathcal{Y}$. This implies that $\operatorname{marg}_X D \supseteq D_{p_X}$ and $\operatorname{marg}_Y D \supseteq D_{p_Y}$ with $p_X := \operatorname{marg}_X p$ and $p_Y := \operatorname{marg}_Y p$, and therefore $D \supseteq D_{p_X} \otimes D_{p_Y} = D_{p_X p_Y}$.

To prove that F belongs to K_D for any D in $\mathcal{D}$, in turn, it suffices to show that F belongs to $K_{D_{p_X p_Y}}$ for every pmf p_X on $\mathcal{X}$ and p_Y on $\mathcal{Y}$. To this end, note that the $p_X p_Y$-expectation of $\sum_{G \in \mathcal{P} \setminus \{G'\}} \mathbb{I}_G[f_{G'} - f_G] + \epsilon$ is

$$\sum_{G \in \mathcal{P} \setminus \{G'\}} E_{p_X}(\mathbb{I}_G)[E_{p_Y}(f_{G'} - f_G)] + \epsilon.$$

Note that not for all G' in $\mathcal{P}$ we have $\sum_{G \in \mathcal{P} \setminus \{G'\}} E_{p_X}(\mathbb{I}_G)[E_{p_Y}(f_{G'} - f_G)] < 0$, because if we had, then also its sum $\sum_{G' \in \mathcal{P}} \sum_{G \in \mathcal{P} \setminus \{G'\}} E_{p_Y}(f_{G'} - f_G) = 0$ must be negative, which is a contradiction. Therefore there is at least one G' in $\mathcal{P}$ such that $\sum_{G \in \mathcal{P} \setminus \{G'\}} E_{p_X}(\mathbb{I}_G)[E_{p_Y}(f_{G'} - f_G)] \geq 0$, and therefore the p-expectation of $\sum_{G \in \mathcal{P} \setminus \{G'\}} \mathbb{I}_G[f_{G'} - f_G] + \epsilon$ is positive, which implies that F belongs to $K_{D_{p_X p_Y}}$, indeed. □